Supplementary Information

A Deterministic Number-Theoretic Framework for the Standard Model Parameter Spectrum

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This document collects the computational artifacts, pre-committed blind-prediction records, and machine-checked theorem inventory for the main paper. Every computation listed here is reproducible from the public repositories (`ugp-physics` and `ugp-lean`); every artifact is SHA-256 certified at the time of its creation. For extended research-programme content beyond this paper's scope — including structural mass relations, topological-signature framing, cryptographic null-discipline tier analyses, and the full Lean-library theorem inventory — see the companion papers listed in § below.

SI Table S1: Computational Artifacts Manifest

Table 1: Computational artifacts cited in the main paper. "SC" = Supplementary Computation. All artifacts reside in `papers/01_SM/canonical_run/` unless otherwise noted. SHA-256 prefixes are over the full artifact file (8 hex characters shown). Full artifact descriptions, per-script invocation commands, and SHA-256 hashes for additional structural-analysis runs are catalogued in `papers/01_SM/PROVENANCE.md` and `papers/01_SM/REPRODUCE.md` in the companion repository.

SC ID	Description	Artifact file	SHA-256 prefix
Core SM predictions (v3)			
SC-A	1000-permutation null suite; canonical UCL fit primary σ falls outside the entire shuffled-null distribution (min permutation $\sigma > 5.6 \times 10^{-2}$ vs canonical baseline $\sigma = 2.95 \times 10^{-5}$)	`nulls_suite_1000.json`	`aec7aad3...`
SC-B	Dual-path mass comparison (UCL 2.3 both arms; theoretical arm: derived renorm$_K$+URC, $\sigma \approx 0.29\%$) plus coefficient-target table vs. Elegant Kernel ($\|\mathbf{\Delta k}\|_\infty \approx 7 \times 10^{-3}$; main text Appendix on verifier modes)	`dual_path_comparison.json`	`47ec8a60...`
SC-D	$\alpha_s(M_Z) = 0.11822$ blind prediction at $+0.24\sigma$ of PDG 2024 ($+0.36\sigma$ of PDG 2022 at pre-commitment time); see pre-commitment trail in Appendix F of main paper	`alpha_s_prediction.json`	`288bac48...`
SC-E	Engine-parameter sensitivity analysis (robustness)	`engine_param_sensitivity.json`	`a805ff7b...`
SC-F	S_3-overlap seesaw sensitivity (neutrino-sector disclosure)	`s3_overlap_seesaw.json`	`76147232...`

SC ID	Description	Artifact file	SHA-256 prefix
SC-G	TE2.2 Residual Classification Conjecture computational certificate (12/34,560 universes pass PSC-Layer-I sieve; all carry SM signature)	`te22_rcc_certificate.json`	`0a4dbb6a...`
SC-H	LOO / k-fold cross-validation methodology for UCL fit; structural CV deviation $\leq$ 1.83% max, 0.96% RMS; all 9 features mutually necessary	`loo_ucl_validation.json`	`5221ba23...`
SC-I	α_w, $\sin^2\theta_W$ gauge-sector validation	`alpha_w_prediction.json`	`2ea93b20...`
SC-J	E_{base} first-principles audit (identifies empirical inputs for engine parameters; Open Problem (ii) in main paper §1.1)	`ebase_first_principles_audit.json`	`10701d35...`
SC-K	Charged-lepton structural anchor $m_e \approx \delta \cdot b_1$ keV at +2.05 ppm ($p = 0.004$ under integer-relation null)	`comp_p01_K*.json`	`08f3c8e4...`
SC-L	Koide S_3 equal-norm $\to$ quadric reformulation; see also SC-GGG	`comp_p01_L*.json`	`741fa438...`
SC-V	Pre-committed blind $m_W = g_2 v/2$ tree-level prediction; missed PDG by $+36\sigma$ (Open Problem (viii); honest falsification)	`comp_p01_V*.json`	`f52b9d33...`
SC-GGG	Koide cyclotomic-12 closed-form numerical verification (m_τ from (m_e, m_μ) at $0.91\sigma_{\text{PDG}}$)	`comp_p01_GGG_koide_closed_form.json`	`143de686...`
SC-S3	Combinatorial-saturation null baseline (transcendental basis saturates at 88 % hit rate at 10 ppm)	`comp_p01_S3*.json`	`148f55a6...`
N_c structural chain (§1.1 contribution 3 and §3.7 of main paper)			
SC-EBF11	Koide angle structural search; discovers a-value pattern $\{1, 9, 5\} = \{N_c^0, N_c^2, (N_c^2 + 1)/2\}$	`comp_p01_EBF_11*.json`	`c262b672...`
SC-EBF12	Top-quark a-value from N_c; structural derivation of $\delta = 7$, $b_1 = 73$	`comp_p01_EBF_12*.json`	`6fd5645b...`
SC-EBF13	strand_count $= (N_c^2 - 1)/4 = 2$; $\theta_{\text{Koide}} = 2/9$ from $N_c = 3$	`comp_p01_EBF_13*.json`	`4d634f57...`
VV from GUT group theory (§3.4 of main paper)			
SC-EBF14	One-loop RGE test of VV coefficient identification (MAP result — confirms N_c identification is algebraic)	`comp_p01_EBF_14*.json`	`c73405fd...`
SC-EBF16	Exact Weyl-dimension-formula verification for A_4 = SU(5) and D_5 = SO(10); structural derivation of α_d, β_d, γ_d	`comp_p01_EBF_16*.json`	`9108c5ac...`

SC ID	Description	Artifact file	SHA-256 prefix
Neutrino structural prediction (§7.3 of main paper)			
SC-EBF18	Discovery of $m_\nu \propto b^{29/9}$ structure; 0.4% mass-squared ratio match	`comp_p01_EBF_18*.json`	`b9e71c1c...`
SC-EBF19	Null test: 8/3654 integer triples $\{b_1 < b_2 < b_3\} \subset [2, 30]$ hit within 1%	`comp_p01_EBF_19*.json`	`a10f2e92...`
SC-EBF21	Three independent structural decompositions of 29/9	`comp_p01_EBF_21*.json`	`71fd4f3a...`
SC-EBF22	Full mechanism: $m_\nu = E_D^2 b^{29/9}/M_{\mathrm{GUT}}$ with $E_D = v_H/29$; $\sum m_\nu = 56$ meV	`comp_p01_EBF_22*.json`	`695f8470...`
SC-EBF24	SO(10) CG analysis; verifies $\dim(126) = 2N_c^2\delta$; third decomposition $(d_{45} - d_{16})/N_c^2 = 29/9$	`comp_p01_EBF_24*.json`	`3ef35f8c...`
EW VEV structural derivation (srrg-lean; grade [A^-], conditional on `psc_ew_entropy_maximization`)			
SC-EBF25	$v_{\mathrm{PSC}} = 246.16$ GeV structural EW VEV derivation via PSC entropy framework; null-discipline saturation 0.35% over 288 formula candidates	`higgs_vev_self_consistent.json`	—
SC-EBF26	Level 1 $m_H = 124.971$ GeV (-2.08σ) using self-consistent EW VEV $v_{\mathrm{self}} = 246.267$ GeV; same two-loop g_2 running as m_W closure	`higgs_mass_level1_closure.json`	—
Referee-closure null-discipline bundles (§4, §5, §7.2.1 of main paper)			
SC-RC1-URC	URC bounded-uniqueness enumeration: 1.6M depth-≤ 3 expressions over UGP-structural atoms; UGP-claimed closures outside top-50; 30 feature-randomisation nulls each finding a better best residual $\Rightarrow$ A/D upheld (Open Problem (iii))	`comp_p01_RC1_urc_bounded_enumeration.json`	`bf5281c3...`
SC-RC1-HL	Higgs $\lambda = \varphi/(4\pi)$ bounded-uniqueness enumeration (UGP atoms + $\pi, 4\pi$): λ-claim at 0.5% outside top-50; null median $\sim 0.0007\%$ $\Rightarrow$ A/D upheld (Open Problem (iv))	`comp_p01_RC1_higgs_lambda_uniqueness.json`	`42a693e5...`
SC-RC1-VV	One-loop SM Yukawa RG test from SO(10) 10 + 126 pattern at M_{GUT}; naive log-linear $Y_d = CY_u^{13/9}Y_l^{-7/6}$ returns $\sim$ 557% max residual (null median $\sim$ 152%); aligns with archival one-loop negatives in the supplementary analysis log; VV coefficients stand as GUT group-theoretic Lean-certified (Open Problem (vii$''$))	`comp_p01_RC1_vv_rg_anomalous_closure.json`	`508f72bd...`

SC ID	Description	Artifact file	SHA-256 prefix
SC-RC1-CKM	CKM UGP-restricted Wilson-coefficient search: 20,000 draws from 13-atom UGP moduli library with Z_6 phases; best max-element residual 40.3%; feature-randomised null reaches 23.8% median (null beats UGP), corroborating the Wilson-coefficient upgrade stress test. Primary QLC-based CKM at 3.3% RMS stands as paper's CKM derivation (Open Problem (v))	`comp_p01_RC1_ckm_ugp_restricted_search.json`	`d4a21842...`
Open-Problem graduated closures (EPIC_038 follow-up; see main paper Open-Problem registry)			
SC-OP-II	Engine-parameter MDL uniqueness enumeration over UGP-structural atoms; certifies Open Problem (ii) closure at A/D	`comp_p01_op_ii_engine_mdl_uniqueness.json`	`b8af23d2...`
SC-OP-II-V	Engine-parameter validation (sensitivity-aware confirmation paired with SC-OP-II)	`comp_p01_op_ii_engine_validation.json`	`c2028753...`
SC-OP-III	URC confinement-scale loop-expansion uniqueness scan over $\Lambda \in [0.2, 0.4]$ GeV; documents partial closure of Open Problem (iii)	`comp_p01_op_iii_urc_confinement.json`	`56f08c56...`
SC-OP-IV	Higgs $\lambda = \varphi/(4\pi)$ MDL uniqueness enumeration over UGP atoms; closes Open Problem (iv) at A/D	`comp_p01_op_iv_higgs_lambda_mdl_uniqueness.json`	`4e1f29d7...`
SC-OP-IV-S	Higgs λ sensitivity scan paired with SC-OP-IV (robustness leg)	`comp_p01_op_iv_higgs_sensitivity.json`	`84dbecb5...`
SC-OP-V	CKM ratio uniqueness analysis under UGP-restricted moduli; companion to SC-RC1-CKM for Open Problem (v)	`comp_p01_op_v_ckm_ratio_uniqueness.json`	`d282bcb2...`
SC-OP-VII	Koide S_3 closed-form uniqueness audit; structural certification supporting Open Problem (vii) closure	`comp_p01_op_vii_koide_s3.json`	`fe9fb5ac...`
SC-OP-VIII	m_W residual decomposition: PDG value sensitivity, SM/PDG tension analysis, ρ-pathway verification, three-loop estimate; documents full closure at -0.42σ (PDG 2024)	`comp_p01_AAA_mw_residual_decomposition.json` + `comp_p01_op_viii_mw_sirlin.json`	`e6b760e7...`

Scope note. This table lists the core artifacts for claims appearing in the main paper. The full set of 24 structural-analysis scripts (`comp_p01_EBF_01` through `comp_p01_EBF_24`) with complete descriptions and SHA-256 hashes is catalogued in `papers/01_SM/PROVENANCE.md`. Step-by-step reproduction commands for all scripts are in `papers/01_SM/REPRODUCE.md`.

SI Table S2: Pre-Committed Blind Predictions and PSC-Calibrated Benchmarks

Table 2: Gauge-coupling results from the Lean-certified bare rationals (g_1^2, g_2^2, g_3^2) = (16/125, 2329/5400, 41 075 281/27 648 000). The two genuinely blind pre-committed predictions (α_s and m_W) are cryptographically verifiable via the Git commit history of the `ugp-lean` and `ugp-physics` repositories; see Appendix F of the main paper for commit hashes, timestamps, and Zenodo DOIs. The $\alpha_{\rm EM}$ row is a PSC-calibrated result (TE_1.P), not a blind prediction; see §5.3 of the main paper. A falsifiable research program has blind predictions on both sides of the success/failure line; this is the beginning of such a pattern.

Observable	UGP prediction	vs. CODATA/PDG	Status
$\alpha_{\rm EM}$ (Thomson, TE_1.P)	7.29737×10^{-3}	+2.39 ppm of CODATA	**PSC-calibrated result** (TE_1.P three-parameter fit anchored on base-137 from UGP $\{0,3,7\}$; bare $g_1^{2,\rm bare}/(4\pi)$ alone gives +0.50%; not a blind chain from bare rationals; SC-I)
$\alpha_s(M_Z)$ (SC-D)	0.11822	$+0.24\sigma$ of PDG 2024 (0.1180±0.0009); $+0.36\sigma$ vs PDG 2022 (0.11790 ± 0.00090)	**Genuinely blind pre-committed prediction** (*consistent-with*, not diagnostic; PDG band $\sim$ 1%; pre-committed in `ugp-lean` 43 days before PDG comparison)
m_W tree-level (SC-V)	80.85 GeV (from $g_2 v/2$, pre-committed)	$+36\sigma$ miss (PDG 80.379 ± 0.012 GeV)	**Clean blind falsification of the naive tree-level pipeline** (Open Problem (viii); honest disclosure)
m_W two-loop + threshold (SC-ZZ)	80.364 GeV (two-loop SM gauge running from the same bare $g_2^{2,\,\rm bare}$, with proper 6→5 flavour threshold matching at m_t)	$\mathbf{-0.42\sigma}$ vs PDG 2024 world average 80.3692 ± 0.0133; -1.28σ vs older PDG 80.379 ± 0.012; 13 MeV gap fully accounted for by SM/PDG W-mass tension (UGP is 9.7 MeV closer to PDG than SM EW fit)	**Closed within PDG** 1σ using PDG 2024 world average (`comp_p01_ZZ_mw_threshold_and_self_consistent.json`, pre-commit SHA-256 `eb4ff13b...`; null test: 500 random $M_2 \in [30, 45]$ GeV, UGP $M_2 = 37.4$ at 52nd percentile; gap decomposition: `comp_p01_AAA_mw_residual_decomposition.json`)
$\Delta m_{21}^2/\Delta m_{31}^2$ (neutrino, SC-EBF18)	0.02936 (Braid Atlas $b = \{5, 11, 19\}$, exponent 29/9)	0.16σ from NuFIT 6.0 (0.02951 ± 0.00098); 0.52% dev; 1.58σ from PDG 2024 world avg; normal hierarchy automatic	**Parameter-free structural prediction** (zero free parameters; see §7.3 of main paper and companion paper [4])

Observable	UGP prediction	vs. CODATA/PDG	Status
$\lambda_H = \varphi/(4\pi)$ (Higgs quartic, SM-18)	0.12876 (MDL-unique at MDL $= 4$ over UGP atoms)	0.26σ from SM tree-level value 0.12928 ± 0.002 (PDG 2024); 0.40% dev	**New confirmed result** (Category $\mathrm{A}_{\mathrm{MDL}}$; MDL-uniqueness certificate; independent of G_F; see §8.5 of main paper)
v_{PSC} (structural EW VEV)	246.16 GeV	-0.024% from $v_{\mathrm{PDG}} =$ 246.22 GeV	**Structural derivation** (grade [A^-]; PSC entropy framework; conditional on 1 named open axiom `psc_ew_entropy_maximization`; zero sorry in srrg-lean; SC-EBF25)
m_H Level 1	124.971 GeV	-2.08σ from PDG 125.20 ± 0.11 GeV	**Level 1 result** (Grade C; uses self-consistent $v_{\mathrm{self}} = 246.267$ GeV from same two-loop g_2 running as m_W closure; 9.1σ bare result is a separate open problem and is not superseded; SC-EBF26)

SI Table S3: Lean Machine-Checked Theorems

The following Lean 4 theorems underpin the Category A structural claims of the main paper. All reside in the `ugp-lean` library [6] with universal axiom signature [`propext`, `Classical.choice`, `Quot.sound`] (standard Mathlib baseline) and zero `sorry` in all tactic proofs.[1] The library comprises 118 modules; see the companion formalization paper [5] for the full theorem inventory; this table is scoped to the core set underpinning this paper's claims. The full theorem inventory is catalogued in the companion formalization paper [5].

Table 3: Core Lean-certified theorems supporting the main paper's Category A claims.

Theorem	Module	Statement
RSUC and Ridge Minimality		
`rsuc_theorem`	`Classification.FormalRSUC`	At $n = 10$, the prime-locked mirror-dual ridge sieve has exactly two survivors $\{(24, 42), (42, 24)\}$; MDL selects the Lepton Seed $(1, 73, 823)$ up to mirror equivalence
`n10_is_minimal_admissible_ridge`	`Compute.SieveBelow10`	$n = 10$ is the smallest admissible ridge level (`native_decide`)
GTE and Quarter-Lock		
`canonical_orbit_triples`	`GTE`	Cascade $(1, 73, 823) \to (9, 42, 1023) \to (5, 275, 65535)$ step-verified

[1] The library additionally contains two analytic number theory lemmas (`dickman_equidistribution_in_APs` and `crt_equidistribution_within_regime`) declared as `axioms` citing Tenenbaum III.6; these do not appear in the axiom closure of any physics theorem in this table. Running `#print axioms` on any theorem below returns exactly the standard Mathlib signature.

Theorem	Module	Statement
quarterLockLaw	QuarterLock	$k_M = k_{\mathrm{gen2}} + \frac{1}{4} k_{L^2}$ (unconditional algebraic identity)
k_L2_eq	ElegantKernel	$k_{L^2} = \delta/2^{n-1} = 7/512$
9/9 UCL Elegant Kernel unconditional closure		
thm_ucl1_fully_unconditional	ElegantKernel.Unconditional	$k_{\mathrm{gen2}} = -\varphi/2$
thm_ucl2_fully_unconditional	ElegantKernel.Unconditional	$k_{\mathrm{gen}} = \varphi\cos(\pi/10)$
thm_ucl4_fully_unconditional	ElegantKernel.Unconditional	k_L closure
thm_ucl5_fully_unconditional	ElegantKernel.Unconditional	k_{const} centering identity
thm_ucl_3	ElegantKernel.MuTriple	Möbius triple $(k_a, k_b, k_c) = (1/8, -3/2, 4/3)$ via Vandermonde uniqueness
Bare gauge couplings (underpinning α_{EM} and α_s predictions)		
g1Sq_bare_eq	Phase4.GaugeCouplings	$g_1^{2,\,\mathrm{bare}} = 16/125$
g2Sq_bare_eq	Phase4.GaugeCouplings	$g_2^{2,\,\mathrm{bare}} = 2329/5400$
g3Sq_bare_eq	Phase4.GaugeCouplings	$g_3^{2,\,\mathrm{bare}} = 41\,075\,281/27\,648\,000$
L_{model} cosmological identity		
L_model_eq_log_residual	LModelDerivation	$L_{\mathrm{model}} = \log_2((D_1 \cdot 5^3)/3)$ from discrete charge $D_1 = 2^4$, golden volume 5^3, orbit length 3
Pentagon–Hexagon Bridge and TT cascade formal closure		
k_gen_pentagon_hexagon_bridge	ElegantKernel.Unconditional.KGenFullClosure	$k_{\mathrm{gen}} + k_{\mathrm{gen2}} = \varphi(\cos(\pi/10) - \cos(\pi/3))$
claim_C_formal	MassRelations.ClaimCBridge	TT mass cascade = SU(3) Weyl rotation at $\pi/6$ per generation with 2^g doubling
pentagon_hexagon_TT_unified_bridge	MassRelations.ClaimCBridge	Five structural facts packaged: pentagon–hexagon bridge, $\beta = \pi/8$, cascade identification
beta_eq_alpha_div_c2_su3	MassRelations.UpLeptonCyclotomic	$\beta = \alpha/C_2(\mathrm{SU}(3)) = (\pi/6)/(4/3) = \pi/8$
N_c structural chain (§1.1 contribution 3 and §3.7 of main paper)		
N_c_determines_everything	MassRelations.KoideAngle	δ, b_1, a_{top}, strand count, and $\theta_{\mathrm{Koide}} = 2/9$ all follow from $N_c = 3$
koide_angle_from_N_c_pure	MassRelations.KoideAngle	$\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ unconditionally from $N_c = 3$ via strand-count identity
delta_from_N_c	MassRelations.KoideAngle	$\delta = N_c + (N_c^2 - 1)/2 = 7$
strand_count_eq_su_nc_adj_div_4	MassRelations.KoideAngle	strand_count $= \dim(\mathrm{SU}(N_c)_{\mathrm{adj}})/4 = (N_c^2 - 1)/4 = 2$
top_a_from_N_c	MassRelations.KoideAngle	$a_{\mathrm{top}} = N_c^4 - a_\tau = 76$
VV down-quark coefficients from GUT group theory (§3.4 of main paper)		

Theorem	Module	Statement
`VV_from_GUT_group_theory`	`MassRelations.DownRational`	$\alpha_d = 1 + \mathrm{rank}(\mathrm{SU}(5))/N_c^2$; $\beta_d = -(1 + Y_{Q_L})$; $\gamma_d = -\dim(45)/\dim(126) = -5/14$
Neutrino seesaw structural closure (§7.3 of main paper)		
`nuSeesawExponent`	`MassRelations.KoideAngle`	$29/9 = N_c + \theta_{\mathrm{Koide}}$
`nu_seesaw_exponent_three_decompositions`	`MassRelations.KoideAngle`	Three independent decompositions: $(N_c^3 + s)/N_c^2 = (4N_c^2 - \delta)/N_c^2 = (d_{45} - d_{16})/N_c^2$
`dim_126_SO10_eq_two_Nc_sq_delta`	`MassRelations.KoideAngle`	$\dim(126_{\mathrm{SO}(10)}) = 2N_c^2\delta = 126$ (cross-sector bridge)
`neutrino_seesaw_structural_closure`	`MassRelations.KoideAngle`	All seesaw structural identities packaged in one conjunction
`delta_b1_is_mersenne_Nc_sq`	`MassRelations.KoideAngle`	$\delta b_1 = 2^{N_c^2} - 1 = 511$ (electron mass factor is the N_c^2-th Mersenne number)
`mersenne_ladder_three_values`	`MassRelations.KoideAngle`	$\delta b_1 = 2^9 - 1$, $c_2 = 2^{10} - 1$, $c_3 = 2^{16} - 1$ (all three Mersenne values certified)
Neutrino mass-squared ratio arithmetic bridge (`MassRelations.NeutrinoMassRatio`; ugp-lean; zero sorry)		
`fn_texture_gives_seesaw_exponent`	`MassRelations.NeutrinoMassRatio`	FN texture $(q_1, q_2) = (3, 2)$ gives $q_1 + q_2/N_c^2 = 3 + 2/9 = 29/9 = \nu_{\mathrm{seesaw}}$
`seesaw_ratio_independent_of_MR`	`MassRelations.NeutrinoMassRatio`	Mass-squared ratio R is independent of common prefactor $C = E_D^2/M_R$ (algebraic)
`neutrino_mass_ratio_coarse_bound`	`MassRelations.NeutrinoMassRatio`	$0.029 < R < 0.030$ where $R = (11^{58/9} - 5^{58/9})/(19^{58/9} - 5^{58/9})$; zero sorry
`neutrino_mass_ratio_tight_bound`	`MassRelations.NeutrinoMassRatio`	$\lvert R - 0.02936\rvert < 0.0001$; proved via unit-width integer bounds on $b^{58/9}$, zero sorry
`neutrino_mass_ratio_within_1pct_of_nufit`	`MassRelations.NeutrinoMassRatio`	$\lvert R - 0.02951\rvert < 0.01 \times 0.02951$; within 1% of NuFIT 6.0 central value; zero sorry
Koide cyclotomic-12 closed form (companion paper [2])		
`cos_sq_pi_div_twelve`	`MassRelations.KoideClosedForm`	$\cos^2(\pi/12) = (2 + \sqrt{3})/4$
`koide_solved_form_root`	`MassRelations.KoideClosedForm`	Positive root $z = 2(x + y) + \sqrt{3}\sqrt{x^2 + 4xy + y^2}$ satisfies the Koide quadratic
`koide_angle_from_gte_structure`	`MassRelations.KoideAngle`	$\theta_{\mathrm{UGP}} = 2/a_\mu = 2/9$ from the muon GTE a-value
Braid Atlas charge theorem (companion paper [3])		
`sm_charge_leptons`	`BraidAtlas.ChargeTheorem`	$Q = W_g/N_c$: leptons ($Q = -1$) and neutrinos ($Q = 0$) certified; quark fractional charges certified via `sm_quarks_fractional_charge`

Theorem	Module	Statement
anomaly_cancellation_forces_Nc_3	BraidAtlas.ChargeTheorem	Anomaly equation $N_c(N_c-3)=0$ forces $N_c=3$ under $N_c>0$; fractional-charge corollary (`nc_eq_3_from_fractional_charge`) gives an independent confirmation
sm_quarks_fractional_charge	BraidAtlas.ChargeTheorem	$N_c \nmid 2$ and $N_c \nmid -1$ certifies fractional quark charges
sm_winding_numbers_from_Nc	BraidAtlas.ChargeDerivation	SM winding set $\{N_c-1,-1,0,-N_c\}$ at $N_c=3$ from $\mathrm{SU}(N_c+2)$ hypercharge bookkeeping
y_ql_unifies_vv_and_winding	BraidAtlas.ChargeDerivation	Hypercharge $Y_{Q_L}=1/(2N_c)$ ties VV slope and braid winding numerator
RCC gauge classification (`PSC.RCCInfiniteFamilies`)		
rcc_all_classical_families	PSC.RCCInfiniteFamilies	Bundled failure modes for infinite classical Lie families B_n, C_n, D_n, A_n at PSC Layers I–II (`BraidAtlas`-adjacent PSC programme)
EW boson c-values and Coxeter-conductor tower law		
ew_boson_c_value_theorem	BraidAtlas.EWBosons	Composite: $c(W)=N_c(N_c+1)-1=11$, $c(Z)=N_c(N_c+1)=12$, $c(H)=N_c(N_c+1)+1=13$; consecutive window; all from $N_c=3$ and GTE orbital constants at $n=10$
ew_c_consecutive_window	BraidAtlas.EWBosons	$c(Z)=c(W)+1$ and $c(H)=c(Z)+1$: unique consecutive triple in GTE orbital constants
ew_c_window_unique	BraidAtlas.EWBosons	$\{c(W),c(Z),c(H)\}=\{11,12,13\}$ is the unique consecutive integer triple drawn from $n=10$ orbital constants
e7_tower_law_complete	BraidAtlas.CoxeterConductorTowerLaw	$\cos(\pi/9)$ irreducible of degree 3; $\varphi(120)=32$; $3\nmid 32 \Rightarrow \mathbb{Q}(\cos(\pi/9)) \not\subset \mathbb{Q}(\zeta_{120})$: E7 Toda mass ratios lie outside the UGP cyclotomic field (falsifier)
EW VEV structural derivation (`SrrgLean.VEVProof` — srrg-lean; grade [A^-], conditional on `psc_ew_entropy_maximizat`		
goldstone_volume_correction_per_generation	SrrgLean.VEVProof.GoldstoneEntropyCorrection	$\varphi^{1/N_{\mathrm{gen}}}$ Goldstone volume correction per generation
psc_entropy_after_contraction	SrrgLean.VEVProof.PSCEntropyDuality	PSC Entropy-Contraction Duality: entropy after contraction determined by PSC extremum

Theorem	Module	Statement
`ew_vacuum_manifold_uniqueness`	`SrrgLean.VEVProof.EWGoldstoneManifold`	S^3 Goldstone manifold uniqueness: 3 massless bosons, $\mathrm{Vol}(S^3) = 2\pi^2$
`srrg_physical_fp_implies_ew_vacuum_manifold`	`SrrgLean.VEVProof.EWVacuumBridge`	Bridge: PhysicalSubspace $\to$ S^3 manifold (conditional on `psc_ew_entropy_maximization`)
`srrg_vev_no_go`	`SrrgLean.FixedPoints.VEVNoGo`	SRRG cannot generate v via dimensional transmutation (no-go theorem)
GTE-P7 dark matter quantum numbers		
`gte_p7_electric_charge_zero`	`BraidAtlas.ChargeTheorem`	$W_{g,\mathrm{mirror}} = 0 \Rightarrow Q_{\text{GTE-P7}} = W_{g,\mathrm{mirror}}/N_c = 0$ (electrically neutral)
`gte_p7_quantum_numbers_neutral`	`BraidAtlas.ChargeTheorem`	Composite: W_g=0, $N_c \mid W_g$; GTE-P7 is SM-neutral (colour-singlet, zero electric charge)
`mirror_triple_residue`	`GTE.GeneralTheorems`	Mirror triple $(1, 73, 2137; g{=}1)$: $\mathrm{mod}(2137, 73) = 20$ matches canonical lepton residue (same algebraic orbit)
`mirror_triple_prime_lock`	`GTE.GeneralTheorems`	$73 \times 29 + 20 = 2137$: mirror triple satisfies the same prime-lock constraint as the canonical lepton triple
S_3-balanced Koide identity (companion paper [2])		
`koide_v1_sq_eq_three`	`MassRelations.KoideAngle`	$\lvert v_1\rvert^2 = (\sum v_g)^2/3 = 3$ for every θ (trivial-representation norm)
`koide_v2_sq_eq_three`	`MassRelations.KoideAngle`	$\lvert v_2\rvert^2 = \sum v_g^2 - \lvert v_1\rvert^2 = 3$ for every θ (standard-2-rep norm)
`koide_equal_norm`	`MassRelations.KoideAngle`	$\lvert v_1\rvert^2 = \lvert v_2\rvert^2 = 3$ for all θ: the S_3-balance condition is a universal ring identity
Mersenne ladder (GTE.MersenneLadder module)		
`c3_phys_formula`	`GTE.MersenneLadder`	$c_3 = 2^{n+2N_c} - 1 = 65535$ connects the Mersenne exponent to the QCD colour rank
`evenStepC_phys_at_n10_Nc3`	`GTE.MersenneLadder`	`evenStepC_phys 10 3 = 65535` verified by `norm_num`
`kprime_is_minimal_double_fib_above_n`	`GTE.MersenneLadder`	$k' = 16 = 2F(6)$ is the unique minimal double Fibonacci number $> n = 10$ (`no_double_fib_between_n_and_kprime`)
`c3_phys_entangled_with_c2`	`GTE.MersenneLadder`	$\gcd(c_2, c_3) = 3 > 1$ (cross-generation entanglement preserved under T_phys)
GTE fiber bundle and orbit stability		
`lepton_orbit_a_values_distinct`	`GTE.FiberBundle`	The canonical lepton orbit a-values $\{1, 9, 5\}$ are all distinct (uniquely identify generation)

Theorem	Module	Statement
canonical_lepton_type_all_generations	GTE.FiberBundle	Canonical lepton fiber type = ChargedLepton across all 3 generations
ugp_residue_is_conserved	GTE.LinearResponse	Zero-mode $c \bmod b = 15$ conserved at all ridge hits
fib_transverse_decay_sq	GTE.LinearResponse	$\psi^2 < 1$: transverse perturbations to the canonical orbit decay (orbit is stable RG fixed point)
Null-discipline: Algebraic Saturation Barrier		
urc_basis_is_saturated	UgpPhysicsLean.NullDiscipline.SaturationBarrier	The URC basis of 1,596,939 algebraic expressions has satDensity > 0.01 (from $e^{319} \geq 320$; zero sorry)
vv_formula_passes_gate	UgpPhysicsLean.NullDiscipline.SaturationBarrier	VV functional form: null density $10^{-5} < 1\%$ — passes saturation gate
vv_coefficients_fail_gate	UgpPhysicsLean.NullDiscipline.SaturationBarrier	VV coefficient values: $54.3\% > 1\%$ null rate — fail gate (Class C, not theorem-grade)
Null-discipline: Theorem Eligibility Criterion (TEC)		
IsTheoremEligible	UgpPhysicsLean.NullDiscipline.TheoremEligibility	Four-gate predicate: PassesNullGate $\wedge$ StabilityGate $\wedge$ PredictiveGate $\wedge$ ConnectiveGate
vv_formula_is_theorem_eligible	UgpPhysicsLean.NullDiscipline.TheoremEligibility	VV functional form passes all 4 TEC gates (theorem-grade)
vv_coefficients_are_classC	UgpPhysicsLean.NullDiscipline.TheoremEligibility	VV coefficient values: 54.3% null rate $\Rightarrow$ Class C (not theorem-grade)
charge_winding_is_theorem_grade	UgpPhysicsLean.NullDiscipline.TheoremEligibility	$Q = W_g/N_c$ is theorem-grade (Theorem C-W passes all 4 gates)
classC_and_classE_exclusive	UgpPhysicsLean.NullDiscipline.TheoremEligibility	Class C and Class E are mutually exclusive
Turing universality		
ugp_is_turing_universal	Universality.TuringUniversal	The UGP substrate is Turing universal via a native Rule-110 embedding
CKM θ_{23} structural ratio		
ridge_eq_D1_mul_mersenne	MassRelations.CKMTheta23	$R_n = D_1 \cdot M_{n-4}$ for all $n \geq 4$ where $M_k = 2^k - 1$
ckm_theta23_ratio_at_n10	MassRelations.CKMTheta23	$\tau(R_{10})/D_1 = 15/8$
ckm_theta23_ratio_uniqueness	MassRelations.CKMTheta23	15/8 ratio is unique to $n = 10$ in $[5, 20]$
op_v_ckm_theta23_closure	MassRelations.CKMTheta23	Bundled OP(v) closure certificate
Koide S_3 discrete arithmetic-mean identity		

Theorem	Module	Statement
`lepton_a_arithmetic_mean`	`MassRelations.KoideS3DiscreteIdentities`	$2{\cdot}a_\tau = a_e{+}a_\mu$ (i.e., $2{\cdot}5 = 1{+}9$)
`lepton_a_discrete_S3_identity`	`MassRelations.KoideS3DiscreteIdentities`	Bundled discrete shadow of S_3 balance
PSC three-route capstone (parametric carrier; supports OP(i))		
`ThreeRouteBundle`	`UgpPhysicsLean.PSC.ThreeRouteForcing`	Prop-only carrier for [Gödel + Landauer + Norfleet] forcing routes
`psc_three_route_capstone`	`UgpPhysicsLean.PSC.ThreeRouteForcing`	Conditional capstone iff PSC (parametric, no smuggling)

Companion formalization paper. The full zero-sorry `ugp-lean` library — including the structural mass relations (TT cyclotomic up-to-lepton identity; $\alpha = \pi/6$ from A_2 Weyl geometry; $\beta = \pi/8$ from Cartan-torus potential; Froggatt-Nielsen UV completion; VV down-type log-linear relation; N_c structural chain; VV GUT group-theory derivation; neutrino seesaw structural closure; braid charge derivations (`BraidAtlas.ChargeDerivation`); PSC RCC classification layer (`PSC.RCCInfiniteFamilies`)), the Koide Newton-step S_3-equivariant flow, and the end-to-end `PhysicalMasses` bridge — is published as an independent formal-methods contribution [5]. The full theorem inventory is in that companion paper; this SI Table S3 is the subset that directly underpins this paper's claims.

Companion Papers

Several distinct research contributions related to this paper are published separately to keep each manuscript focused:

- **Koide Cyclotomic-12 Closed Form and N_c Structural Derivation** [2]: standalone paper presenting (i) the Koide closed form and its cyclotomic-12 identification as elementary facts about the charged-lepton mass spectrum (independent of UGP), (ii) the structural derivation of the Koide phase $\theta = (N_c^2 - 1)/(4N_c^2) = 2/9$ from the QCD colour rank $N_c = 3$ (Lean-certified), and (iii) the connection to the neutrino seesaw exponent 29/9. Full derivation, S_3-equivariant Newton-step dynamical realisation, and Lean-formalisation details. Source: `papers/18_koide_cyclotomic/`.

- **Cyclotomic-12 Mass Structure** [1]: companion paper presenting the TT and VV structural mass relations, A_2 Weyl-chamber geometric origin of $\alpha = \pi/6$, Froggatt-Nielsen UV completion, VV coefficient structural derivation from SU(5)/SO(10) group theory, and three null-discipline tests on the VV down-sector coefficients. Together with the main paper and the Koide companion, this determines all nine charged-fermion masses from two scalar inputs. Source: `papers/19_cyclotomic_mass_structure/`.

- **Neutrino Mass-Squared Ratio from the Braid Atlas** [4]: companion paper presenting the full derivation of $\Delta m^2_{21}/\Delta m^2_{31} = 0.02936$ from Braid Atlas b-values $\{5, 11, 19\}$ with exponent 29/9; three independent structural decompositions; Dirac scale $E_D = v_H/29$; null tests and preregistered falsification criteria for JUNO, DUNE, CMB-S4, and LEGEND-1000. Source: `papers/22_neutrino_masses/`.

- **Braid Atlas** [3]: the universal arithmetic-to-topology dictionary Ψ mapping GTE triples to braid-group elements, with theorems deriving SM quantum numbers (family, generation, spin, charge) from braid topological invariants. Source of the right-handed neutrino b-values $\{5, 11, 19\}$.

- **UGP formal-methods paper** [5]: the `ugp-lean` library as a stand-alone formal-methods contribution, with the complete zero-sorry theorem inventory, zero-`sorry` accounting, and detailed build/reproducibility instructions.

Reproduction

Every artifact in SI Table S1 can be regenerated by running the corresponding Python script in `papers/01_SM/canonical_run/` of the public `ugp-physics` repository. All scripts are self-contained (dependencies: Python 3.9+, NumPy, and optionally SciPy/SymPy) and produce deterministic output. The primary verifier (all SM predictions) is invoked as:

```
cd UGP_GTE_SM_Verifier
python3 UGP_GTE_SM_Verifier.py \
  --preset-fullstack --n 10 --full-derivation 1
```

The structural-analysis scripts (SC-EBF11 through SC-EBF24) are each independently invocable from `papers/01_SM/canonical_run/`; step-by-step commands are in `papers/01_SM/REPRODUCE.md`. SHA-256 of each output JSON must match the prefix listed in Table S1 (timestamps in JSON payloads may differ; SHA-256 is computed over the prediction block for blind tests).

Every Lean theorem in SI Table S3 can be verified by:

```
git clone https://github.com/novaspivack/ugp-lean
cd ugp-lean
lake build
```

A clean build completes with zero `sorry` warnings and the standard Mathlib axiom signature. The specific commit supporting the α_s blind prediction is `27e6823` (timestamp: 2026-03-05 14:09:28 EST). The pre-commitment trail for the 43-day gap is documented in Appendix F of the main paper.

Scope note. This Supplementary Information document is scoped to the claims of the main paper. Additional structural results (TT/VV mass relations, the Tier I/II/III null framework, the Braid Atlas topological signature, and the m_W SM-machinery recovery analyses) are documented in the companion papers listed above. The SC-RC1 artifact group (SC-RC1-URC, SC-RC1-HL, SC-RC1-VV, SC-RC1-CKM) in Table S1 bundles null-disciplined computations supporting the Open-Problem disclosures in §4, §5, and §7.2.1 of the main paper: they *confirm* the A/D status of the URC closures, the Higgs λ identification, the VV-Lagrangian open item, and the CKM Wilson-coefficient upgrade pathway, rather than promoting any of them to Category A. This upholds the existing paper classifications with explicit computational backing instead of verbal hedges.

References

[1] N. Spivack. Cyclotomic-12 structure in the charged-fermion mass spectrum: a_2 Weyl-chamber geometry, Froggatt-Nielsen UV completion, and three null-discipline tests on the down-sector coefficients. Zenodo, 2026. Paper P19 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168805.

[2] N. Spivack. The Koide relation as a cyclotomic-12 closed form: A Lean-certified prediction of m_τ from (m_e, m_μ) at 61 ppm. Zenodo, 2026. Paper P18 in the UGP Physics series. https://doi.org/10.5281/zenodo.20168795.

[3] Nova Spivack. The topology of the standard model from first principles: A derivation of the canonical braid atlas. Zenodo, 2025. https://doi.org/10.5281/zenodo.20169984.

[4] Nova Spivack. Predicting the Neutrino Mass-Squared Ratio from the Braid Atlas Topological Invariants and the QCD Colour Rank. Zenodo, 2026. Paper P21 in the UGP Physics series. https://doi.org/10.5281/zenodo.20170120.

[5] Nova Spivack. ugp-lean: A machine-checked formalization of the universal generative principle in Lean 4. Zenodo, 2026. Formalization paper (latest version): https://doi.org/10.5281/zenodo.19554688. Concept DOI (always-latest): https://doi.org/10.5281/zenodo.19433538. 117 modules across twelve

architectural layers including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (ChargeTheorem, CompositeTriples, ChiralitySquaring, ChargeDerivation, CoxeterConductor, CoxeterConductorTowerLaw, EWBosons), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. Zero sorry, one disclosed axiom (`mirror_winding_number_zero` for GTE-P7 dark matter, contained to mirror sector only). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

[6] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. https://doi.org/10.5281/zenodo.19554700. 117-module Lean 4 library (module count updated as of 2026-05-09). `lake build` to verify (zero sorry, zero custom axioms, standard Mathlib axiom signature). Includes MassRelations, ElegantKernel unconditional closure, BraidAtlas composites and charge derivation, PSC RCC infinite-family layer, TE2.2 scan certification. New in GTE.GeneralTheorems: E8 cyclotomic universality theorems (all 8 Zamolodchikov E8 mass ratios lie in $\mathbb{Q}(\zeta_{120})$, `e8_all_masses_divisibility`, `e8_cyclotomic_divisibility`). Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.